

Siyan(Simon) Wen

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Portfolio: <https://siyanwen.github.io/>

Professional Summary

Seasoned software engineer with hands-on experience in backend and distributed systems, specializing in **Java, Python, Go**, and **cloud-native microservices**. Led a machine learning team to third place in a competition by implementing K-means clustering and feature reduction pipelines. Built **Spring Boot** based event driven e-commerce backend and REST APIs, leveraging **containerization** and **CI/CD** to streamline workflows.

Passionate about developing scalable, reliable platform components.

Technical Skills

- **Languages:** java, C, C++, C#, Python, Kotlin, Go, JavaScript, TypeScript, PHP, Bash, Ocaml
- **Messaging & Databases:** Kafka, MySQL, PostgreSQL, Cassandra, MongoDB, Redis, Elasticsearch, SQL Databases
- **Backend:** Spring Boot, Spring Security, Resilience4j, Django, Flask, Node.js, Express.js, Maven, Gradle, Postman, Swagger
- **Frontend:** React 18, TypeScript, HTML5, CSS3, AJAX
- **Mobile:** Android Studio, Ktor, Jetpack Compose, Google Exoplayer, Room
- **Microservices:** RESTful APIs, gRPC, GraphQL, Event-driven Architecture, Saga Pattern, CQRS, API Gateway, Service Mesh, Circuit Breaker
- **Architecture & Patterns:** Singleton, Factory, Builder, MVVM, MVC, DAO, DTO
- **Cloud & DevOps:** AWS (EC2, S3, RDS, EKS, ELB, IAM), GCP, Docker, containerization, Kubernetes, Helm, Terraform, Jenkins, GitHub Actions, CI/CD
- **Testing:** JUnit, Mockito, JaCoCo, Testcontainers
- **Development Practices & Tools:** Jira, Trello, Software Development Best Practices, Infrastructure as Code (IaC), Git
- **Monitoring:** Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), OpenTelemetry, Jaeger, Spring Actuator, DataDog, NewRelic
- **Methodologies:** Agile (Scrum), TDD, BDD, DevOps, Full SDLC
- **AI: ML:** Regression, classification, clustering, KNN, SVM, Decision Tree; **NLP:** word2vec, Skip-Gram, Word Embedding, Language Modeling, N-Gram, RNN, LLM, attention, transformer, seq2seq model, dependency parsing, CNN, GAN, RAG; **RL:** UCS, A*, Markov Decision Process, Monte Carlo, SARSA, Q-learning, policy gradient, HMM
- **AI Tools:** Hugging Face, Langchain, Elasticsearch, Tensorflow, keras, scikit-learn, Numpy, Pandas

Projects

Social media with AI image generation | [Front End](#) [Back End](#)

- Designed a scalable social platform for generating and sharing AI-created images, integrated multi-threading to support high concurrency and reliability
- Built a Go backend with **JWT authentication**, containerized with Docker, and deployed via GitHub Actions CI/CD on **Google Cloud App Engine**
- Integrated OpenAI's **DALL·E 3** for prompt-based image generation.
- Implemented **Elasticsearch** for keyword and user-based post search.
- Managed image/video uploads with **Google Cloud Storage** buckets.
- Technologies: **Go, React.js, DALL·E 3, GCP (App Engine, GCS, GCE), Elasticsearch**

PDF AI Agent | [Front End](#)

- Built a chat bot that supports PDF upload. By integrating **Retrieval-Augmented Generation (RAG)** technology, the results generated by LLM could be more reliable, domain specific and avoid hallucination.
- Engineered an interactive conversational UI of PDF uploader and input bar, enabling users to effortlessly upload and interact with PDF documents in real-time.
- Architected and implemented RESTful APIs via **Express** and **Node.js**, optimized for high-performance request handling, integrated unit tests with Jest, and conducted code reviews to ensure reliability
- Integrated **OpenAI's GPT-3.5 Turbo** and **Langchain** to load, divide, store, retrieve, and output documents.
- Used a vectorstore to cache generated embeddings in memory for efficient retrieval, enhancing scalability with memory-optimized data structures
- Integrated voice chat mode by integrating speech-to-text and text-to-speech technologies.
- Technologies: **Express.js, GPT-3.5 Turbo, Langchain, React.js, RAG**

Food delivery platform | [Frontend + backend](#)

- Developed a full-stack microservice e-commerce platform using **Spring Boot 3** and **React.js**, comprising 5 domain services: Account, Inventory, Cart, Order and Payment.
- Applied **polyglot persistence** across services - PostgreSQL (JPA) for accounts and payments, MongoDB for product catalog, and Apache Cassandra for order storage - leveraging each database's strengths for its domain.
- Implemented **event-driven** order processing with **Apache Kafka** (KRaft mode): OrderService publishes order events and

consumes payment confirmation events, decoupling the payment lifecycle via an asynchronous **Saga** pattern.

- Leveraged **Spring Cloud OpenFeign** for synchronous inter-service communication with **Resilience4j** circuit breakers and fallback factories, preventing cascade failures across service boundaries.
- Designed a **CQRS** pattern in OrderService using dual denoted Cassandra tables (orders_by_id, orders_by_user) to optimize for distinct query access patterns without joins.
- Extracted a shared common-security library providing JWT authentication filters, HMAC-SHA token generation, and a FeignAuthInterceptor for service-to-service token propagation, enforcing consistent auth across all services.
- Built the frontend with **React 18** and **Ant Design**, proxied through the API Gateway, enabling cart management, order placement, and payment workflows.
- Containerized the full stack with Docker Compose including health-check dependencies and a Cassandra init container for schema bootstrapping; enforced test coverage with **JaCoCo** and **Testcontainers** for MongoDB integration tests.
- Technologies: **Java, Spring Boot, Maven, Spring Cloud Gateway, Netflix Eureka, Spring Security, Spring Cloud OpenFeign, Hibernate, Spring JPA, React.js, Ant Design, PostgreSQL, Cassandra, MongoDB, Junit, Mockito, Resilience4j, JaCoCo**

Android Music App | [Front End](#) [Back End](#)

- Created a full-featured music app using Android Studio, Jetpack libraries and MVVM architecture.
- Supported song playback, album browsing, and favorites collection.
- Implemented **MVVM** architecture on Home/Album/Collection UI using **Jetpack Compose**.
- Enabled seamless navigation via Jetpack Navigation Components.
- Used RESTful API to handle requests from json-server by leveraging **Retrofit**.
- Enabled **Google Exoplayer** to handle music playback globally and Room for local caching.
- Technologies: **Kotlin, Ktor, Jetpack Compose, Hilt Dependency Injection, Google Exoplayer, Room**

Professional Experience

3D Print Joy

Jun 2026 - Current

3D Printing Specialist

Ontario, CA

- Responsibilities include coordinating with operations and warehouse teams, producing 3D-printed items for shipment, and enhancing printing efficiency and quality through model optimization, slicing parameter adjustments, and standardized machine operation.
- Managed the maintenance and servicing of over 90 3D printers, reducing the **machine failure rate** from **19%** to **3%**.
- Optimized print scheduling, resulting in a **58%** increase in **daily output**.
- Improved the **yield rate** by **20%** by identifying and addressing the root causes of defects.

Stevens Institute of Technology

Jan 2025 - May 2025

NLP Research Assistant

Hoboken, NJ

- Responsible for presenting the online annotation interface, crawling and backing up webpages, optimizing prompts, and generating training data for Named Entity Recognition (NER) using the OpenAI API.
- Improved efficiency by replacing traditional methods with a Flask-based HTML interface, which streamlined data annotation processes.
- Engineered an exquisite UI that displays paginated annotation questionnaire with result export, timer and auto page turning features. Each page displays a GitHub commit including code block with syntax highlights, code change highlights and interactive file hierarchy. The Flask backend populates the frontend with a spreadsheet using Jinja.
- Developed and debugged a BeautifulSoup web crawler to back up website resources.

Suzhou Qudou Culture Media Co. Ltd.

Feb 2022 - Apr 2022

Character Animator

Jiangsu, China

- Engaged in the animation layout, Special Effects illustration and camera movement design of ep10 and ep14, the most challenging parts (fighting scenes) of Westward using Maya 2018.
- Polished the motion data of the motion capture device, produced the initial expressions, and determined the rhythm and position.
- Improved work efficiency and the quality of the delivered animations by taking the lead in using the Maya animation special effects plugin.

Shenzhen Liqian Technology Co. Ltd.

Oct 2019 - Mar 2021

E-commerce Operations Specialist

Shenzhen, China

- Responsible for Amazon store operations (product listing, promotions, shipping, restocking, FBA, and after-sales service).
- Identified high-potential products and optimized supply chains for Amazon storefronts.
- Conducted Search Engine Optimization (SEO) and created promotional strategies to boost sales.
- Improved product page clarity with detailed specs and model guides, cutting return rates by 80%.

Education

Stevens Institute of Technology

Sep 2023 - May 2025

Master of Science, Computer Science

Hoboken, NJ

Central Academy of Fine Arts

Sep 2014 - Jun 2019

Bachelor's, Architectural Design

Beijing, China